$$(\Lambda\alpha^k.v)[\tau] \quad \rightarrow \quad v[\alpha := \tau] \qquad \text{Step-E-TApp}$$

$$\texttt{handler}^{\varepsilon}_{m}(h)\ v \quad \rightarrow \quad \texttt{handle}^{\varepsilon}_{m}(h, v\ ()) \qquad \text{Step-E-Handler}$$

$$\texttt{handle}^{\varepsilon}_{m}(h, v) \quad \rightarrow \quad v \qquad \text{Step-E-Return}$$

$$\frac{\tau = \tau_1 \rightarrow^{\varepsilon} \tau_2}{(\mu x : \tau.v_1)\ v_2 \rightarrow v_1[x := \mu x : \tau.v_1]\ v_2} \text{ Step-E-Fix}$$

$$\frac{\tau_1 = \forall\alpha^k.\tau_3}{(\mu x : \tau_1.v)[\tau_2] \rightarrow v[x := \mu x : \tau_1.v][\tau_2]} \text{ Step-E-TApp-Fix}$$

$$\frac{\begin{array}{c}\mathsf{op} \notin \texttt{bop}(E) \wedge (\mathsf{op} \rightarrow f) \in h \qquad \mathsf{op} : \forall\alpha.\tau_1 \rightarrow \tau_2 \in \Sigma(l) \\ k = \lambda^{\varepsilon} x : \tau_2[\alpha := \tau].\texttt{handle}^{\varepsilon}_{d}(h, E[x])\end{array}}{\texttt{handle}^{\varepsilon}_{d}(h, E[\texttt{perform}\ \mathsf{op}\ \tau\ v]) \quad \rightarrow \quad f[\tau]\ v\ k} \text{ Step-E-Perform-Deep}$$

$$\frac{\begin{array}{c}\mathsf{op} \notin \texttt{bop}(E) \wedge (\mathsf{op} \rightarrow f) \in h \qquad \mathsf{op} : \forall\alpha.\tau_1 \rightarrow \tau_2 \in \Sigma(l) \\ k = \lambda^{\langle l,\ \varepsilon\rangle} x : \tau_2[\alpha := \tau].E[x]\end{array}}{\texttt{handle}^{\varepsilon}_{s}(h, E[\texttt{perform}\ \mathsf{op}\ \tau\ v]) \quad \rightarrow \quad f[\tau]\ v\ k} \text{ Step-E-Perform-Shallow}$$

$$\frac{E \neq \square}{E\left[\texttt{error}_p\right] \Longrightarrow \texttt{error}_p} \text{ Step-E-Error}$$

$$\frac{e_1 \rightarrow e_2}{E[e_1] \Longrightarrow E[e_2]} \text{ Step-E} \qquad \frac{e_1 \dashrightarrow e_2}{E[e_1] \Longrightarrow E[e_2]} \text{ Step-E-Base}$$

Helper functions

$$\begin{array}{lcl}
\texttt{bop}(\square) & = & \varnothing \\
\texttt{bop}(E\ e) & = & \texttt{bop}(E) \\
\vdots & \vdots & \vdots \\
\texttt{bop}(\texttt{handle}^{\varepsilon}_{m}(h, E)) & = & \texttt{bop}(E) \cup \{\mathsf{op} \mid (\mathsf{op} \rightarrow f) \in h\}
\end{array}$$

Listing 1: Reduction Rules for `Effects`